Floating point: denary to binary 4

Advanced: Challenge 2

Select the correct **binary** number representation of the decimal number **-16.75_{10}**.

The binary number representation used is a **normalised floating point number** with an 8-bit mantissa and a 4-bit exponent. The mantissa and exponent are both stored using two's complement.

mantissa	exponent
0.1000011	0110

mantissa	exponent
1.0111101	0101

mantissa	exponent
1.0111101	1011

mantissa	exponent
101111.01	0000

Quiz:
STEM SMART 2026 Residential Extend Quiz 1 - Floating Point Numbers

Floating point: denary to binary 3

Advanced: Challenge 2

Ayeesha is flying home to see her family for the festive season. She has just learnt about floating point representation and has decided to store the cost of her plane ticket in her budget as a floating point binary number. Her plane ticket costs £234.75.

Convert this figure to a normalised two's complement binary floating point number using a 12-bit mantissa and 6-bit exponent. Which is the correct answer?

mantissa	exponent
0.01110101011	001001

mantissa	exponent
1.11010101100	000111

mantissa	exponent
0.11101010110	001000

mantissa	exponent
1.00010101010	001001

Quiz:
STEM SMART 2026 Residential Extend Quiz 1 - Floating Point Numbers

Floating point: binary to denary 2

Advanced: Challenge 2

The binary number shown below is represented as a **normalised floating point number** with a 10-bit mantissa and a 4-bit exponent. The mantissa and exponent are both stored using two's complement.

mantissa												**exponent**			
1	.	0	0	0	0	1	0	1	0	0		0	1	0	1

Convert the floating point number into denary.

Type your answer as a **signed decimal number** (e.g. $+3.25$) - do not leave any spaces in your answer.

Quiz:
STEM SMART 2026 Residential Extend Quiz 1 - Floating Point Numbers

Floating point: binary to denary 4

Advanced: Challenge 2

The binary number shown below is represented as a **normalised floating point number** with an 8-bit mantissa and a 4-bit exponent. The mantissa and exponent are both stored using two's complement.

mantissa										exponent			
1	.	0	0	1	0	1	0	0		1	1	1	1

Which of the following options shows the correct denary representation of the number?

- $+0.578125$
- -0.421875
- -1.6875
- -27648

Quiz:
STEM SMART 2026 Residential Extend Quiz 1 - Floating Point Numbers

Floating point: number range 3

Advanced: Challenge 2

A floating point number is represented by a mantissa and an exponent. Both parts are stored using two's complement representation. The floating point number is normalised.

The number of bits allocated to the **mantissa** affects the **precision** of numbers that can be represented, while the **exponent** affects the **range** of the numbers.

If 5 bits are allocated to the mantissa and 3 bits are allocated to the exponent, what is the **lowest (most negative) number** that can be represented?

Give your answer as a denary (base-10) number.

Quiz:
STEM SMART 2026 Residential Extend Quiz 1 - Floating Point Numbers

Floating point: range and precision

Advanced: Challenge 2

A floating point number is represented by a mantissa and an exponent.

If 16 bits are allocated to the floating point number:

- The bits could be split equally between the mantissa and the exponent
- The mantissa could have more bits than the exponent
- The exponent could have more bits than the mantissa

What are the **two** effects of having a **larger exponent** and a **smaller mantissa** on the **range and precision** of the numbers that can be represented?

- [ ] The precision of the numbers that can be represented is decreased.
- [ ] The precision of the numbers that can be represented is increased.
- [ ] The range of numbers that can be represented is increased.
- [ ] The range of numbers that can be represented is decreased.

Quiz:
STEM SMART 2026 Residential Extend Quiz 1 - Floating Point Numbers

Identify the mantissa and exponent 3

Advanced: Practice 2

A real number can be represented in floating point form using the mantissa, number base, and exponent.

Part A

Identify the **mantissa** in the following floating point denary number:

0.65431×10^5

Part B

Identify the **exponent** in the following floating point denary number:

0.65431×10^5

0.65431×10^5 is **not** written in standard form (scientific notation). In standard form, the mantissa of a denary number must always lie in the range 1 to 10 (>=1 and <10), so the number would be written as 6.5431×10^4.

Quiz:
STEM SMART 2026 Residential Extend Quiz 1 - Floating Point Numbers

Limitations of floating point systems

Advanced: Challenge 1

LLM marked question

30 attempts remaining today

In this question, a 7-bit mantissa and 5-bit exponent is used, both of which are stored using two's complement.

A round error occurs when the decimal value of 26.75 in converted into binary using a floating-point system.

Explain **why** this occurs and what might be the **result** of this happening.

[2 marks]

Quiz:
STEM SMART 2026 Residential Extend Quiz 1 - Floating Point Numbers